

Master Cisco Commands for Better Network Configuration

Command reference with explanations, benefits, and current best-practice notes

Why this is done / Benefit: This guide explains what each configuration block accomplishes and why an administrator may use it.

Best-practice note: Several source commands use weak example passwords, Telnet, older VTP behavior, 1024-bit RSA keys, or legacy interface names. Treat them as lab examples. In production, use unique secrets, enable secret, SSH only, AAA, modern key sizes, and platform-supported syntax.

1. Basic Router Configuration

Hostname and Basic Security

```
Router(config)#hostname R1
R1(config)#enable secret cisco
R1(config)#service password-encryption
R1(config)#banner motd $
UNAUTHORIZED ACCESS IS PROHIBITED
$
```

Why this is done / Benefit: Assigning a hostname makes the device easy to identify in prompts, logs, monitoring systems, and troubleshooting output. The enable secret protects privileged EXEC mode, password encryption prevents basic clear-text display in the running configuration, and a login banner establishes an authorized-use warning.

Best-practice note: Do not use “cisco” as a real password. Prefer a long unique secret, centralized AAA, and stronger password hashing supported by the platform. “service password-encryption” is obfuscation, not strong encryption.

Console, VTY, and AUX Lines

```
R1(config)#line con 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 30 0
R1(config)#line vty 0 4
R1(config-line)#password cisco
R1(config-line)#login
```

```
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 30 0
R1(config)#line aux 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#logging synchronous
R1(config-line)#exec-timeout 30 0
```

Why this is done / Benefit: Line configuration controls local console, remote virtual terminal, and auxiliary access. Login enforcement requires authentication, logging synchronous keeps system messages from disrupting commands being typed, and an idle timeout closes abandoned sessions to reduce unauthorized access and resource consumption.

Best-practice note: Use “login local” or AAA instead of a shared line password. Disable the AUX line when unused. Remote access should be SSH only.

Aliases and SSH

```
R1(config)#alias exec c configure terminal
R1(config)#alias exec s show ip interface brief
R1(config)#alias exec sr show running-config
R1(config)#no ip domain-lookup
R1(config)#ip domain-name example.com
R1(config)#username admin password cisco
R1(config)#crypto key generate rsa
R1(config)#ip ssh version 2
R1(config)#line vty 0 4
R1(config-line)#login local
R1(config-line)#transport input telnet ssh
```

Why this is done / Benefit: Aliases speed up repetitive administration and reduce typing errors. Disabling domain lookup prevents mistyped commands from causing long DNS lookup delays. A domain name, local user, RSA key pair, and SSH version 2 provide encrypted remote management and device authentication.

Best-practice note: Use “username admin secret ...” rather than “password,” generate at least a 2048-bit RSA key where supported, and configure “transport input ssh” to block Telnet.

2. Static Routes, RIP, and OSPF Setup

Static and Default Routes

```
R1(config)#ip route 10.1.2.0 255.255.255.0 10.1.128.1
R1(config)#ip route 10.1.2.0 255.255.255.0 Serial 0/0
R1(config)#ip route 0.0.0.0 0.0.0.0 199.1.1.1
```

Why this is done / Benefit: Static routes provide predictable forwarding to specific networks without routing-protocol overhead. A default route sends traffic for unknown destinations toward an upstream router, simplifying edge-router tables. These routes are useful for small, stable networks, backup paths, and controlled traffic flow.

Best-practice note: On multi-access Ethernet links, specifying a next-hop IP is generally clearer and avoids unnecessary recursive or ARP behavior. Confirm the destination, next hop, and administrative distance before deployment.

RIP Configuration

```
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 10.0.0.0
R1(config-router)#no auto-summary
R1(config-router)#passive-interface serial 0/0
```

Why this is done / Benefit: RIPv2 dynamically exchanges routes and supports classless subnet information. Disabling automatic summarization prevents incorrect classful summaries across discontinuous networks. A passive interface advertises its connected network without sending routing updates out that interface, reducing unnecessary traffic and exposure.

Best-practice note: RIP is primarily suitable for labs and small legacy networks because of its 15-hop limit and slow convergence. OSPF or another modern routing protocol is usually preferred in production.

RIP Verification

```
R1#show ip protocols
R1#show ip route
R1#show ip route rip
R1#show ip route 10.1.1.1
```

Why this is done / Benefit: These commands confirm which routing protocols are active, what networks are being advertised, which routes entered the routing table through RIP, and how the router resolves a particular destination. Verification catches missing networks, wrong masks, passive-interface mistakes, and route-selection problems.

OSPF Configuration

```
R1(config)#router ospf 10
R1(config-router)#network 10.0.0.0 0.255.255.255 area 0
R1(config-router)#network 172.16.8.0 0.0.7.255 area 0
```

```
R1(config-router)#network 192.168.1.254 0.0.0.0 area 1
R1(config-router)#router-id 1.1.1.1
```

Why this is done / Benefit: OSPF provides fast, scalable, loop-free dynamic routing. Network statements select participating interfaces and assign them to areas. Area 0 forms the backbone, while additional areas reduce link-state database size and update scope. A stable router ID makes neighbor relationships and troubleshooting more predictable.

Best-practice note: Wildcard masks must match the intended interfaces. In many environments, interface-level OSPF commands are easier to audit than broad network statements.

OSPF Tuning and Security

```
R1(config)#interface loopback 0
R1(config-if)#ip address 1.1.1.1 255.255.255.255
R1(config-if)#ip ospf hello-interval 2
R1(config-if)#ip ospf dead-interval 6
R1(config-if)#ip ospf cost 55
R1(config-if)#bandwidth 128
R1(config-router)#auto-cost reference-bandwidth 1000
R1(config-if)#ip ospf authentication message-digest
R1(config-if)#ip ospf message-digest-key 1 md5 cisco
R1(config-router)#maximum-paths 6
R1(config-router)#passive-interface serial 0/0
```

Why this is done / Benefit: A loopback creates a stable logical address that remains up independently of physical links and is commonly used as a router ID. Hello/dead timers control neighbor failure detection, cost and bandwidth influence best-path calculation, reference bandwidth prevents fast links from receiving identical default costs, authentication limits unauthorized neighbor formation, equal-cost multipath increases path utilization and resilience, and passive interfaces reduce unnecessary adjacencies.

Best-practice note: OSPF timers must match on neighboring routers. MD5 is legacy; use stronger authentication supported by your IOS release. Set the same reference bandwidth consistently on every OSPF router.

3. Basic Switch Configuration and SSH Access

Initial Setup and Security

```
SW1(config)#hostname SW1
SW1(config)#enable secret cisco
SW1(config)#enable password notcisco
SW1(config)#service password-encryption
SW1(config)#banner motd $
```

```
UNAUTHORIZED ACCESS IS PROHIBITED
$
```

Why this is done / Benefit: The hostname identifies the switch, the enable secret protects privileged access, password obfuscation reduces casual exposure in configuration output, and the warning banner communicates access policy. These basics make administration clearer and reduce accidental or unauthorized privileged access.

Best-practice note: The enable secret takes precedence over enable password, so configuring both is usually unnecessary. Use only a strong enable secret and centralized AAA where possible.

Line Configuration

```
SW1(config)#line con 0
SW1(config-line)#password cisco
SW1(config-line)#login
SW1(config)#line vty 0 4
SW1(config-line)#password cisco
SW1(config-line)#login
SW1(config-line)#history size 15
SW1(config-line)#exec-timeout 10 30
SW1(config-line)#logging synchronous
```

Why this is done / Benefit: This secures console and remote sessions, retains a useful command history, automatically closes idle VTY sessions after 10 minutes and 30 seconds, and keeps asynchronous log messages from corrupting the command line. The result is more secure and usable device management.

Best-practice note: Prefer local or AAA authentication and SSH rather than shared VTY passwords.

Management Interface and Gateway

```
SW1(config)#interface vlan 1
SW1(config-if)#ip address 172.16.1.11 255.255.255.0
SW1(config-if)#no shutdown
SW1(config)#ip default-gateway 172.16.1.1
SW1#copy running-config startup-config
SW1#write
```

Why this is done / Benefit: The switch virtual interface gives a Layer 2 switch an IP address for remote management. The default gateway allows management traffic to reach administrators outside the local subnet. Saving the running configuration preserves changes after a reboot or power loss.

Best-practice note: Use a dedicated management VLAN instead of VLAN 1. “copy running-config

startup-config” is the preferred explicit save command; “write” is a legacy shortcut.

SSH Configuration

```
SW1(config)#ip domain-name example.com
SW1(config)#username admin password cisco
SW1(config)#crypto key generate rsa
How many bits in the modulus [512]: 1024
SW1(config)#ip ssh version 2
SW1(config)#line vty 0 4
SW1(config-line)#login local
SW1(config-line)#transport input telnet ssh
```

Why this is done / Benefit: This creates the identity and cryptographic material required for encrypted remote administration. SSH version 2 protects credentials and commands from interception, while local login ties access to named user accounts rather than a shared line password.

Best-practice note: Use “username ... secret,” at least 2048-bit keys where supported, and “transport input ssh” so Telnet is not accepted.

4. VLANs, Port Security, and Verification

Shortcuts and Interface Settings

```
SW1(config)#alias exec c configure terminal
SW1(config)#alias exec s show ip interface brief
SW1(config)#alias exec sr show running-config
SW1(config)#interface fastEthernet 0/1
SW1(config-if)#description LINK TO INTERNET ROUTER
SW1(config-if)#speed 100
SW1(config-if)#duplex full
SW1(config-if)#interface range fastEthernet 0/5 - 10
SW1(config-if-range)#duplex full
```

Why this is done / Benefit: Aliases improve efficiency. Interface descriptions document physical or logical purpose directly in the configuration. Explicit speed and duplex settings can solve negotiation problems with legacy devices, and interface ranges apply a consistent setting to multiple ports, reducing repetitive work and configuration drift.

Best-practice note: Leave speed and duplex on auto unless a connected device requires matching manual settings. A mismatch can cause severe performance problems. Modern platforms may use GigabitEthernet or other interface names.

Verification Commands

```
SW1#show version
SW1#show running-config
SW1#show startup-config
SW1#show history
SW1#show ip interface brief
SW1#show interface vlan 1
SW1#show interfaces description
SW1#show interfaces status
SW1#show crypto key mypubkey rsa
SW1#show dhcp lease
```

Why this is done / Benefit: These commands validate software and hardware details, compare active and saved configuration, review recent commands, check interface addressing and operational state, confirm interface documentation and link status, verify SSH key material, and inspect DHCP-derived management settings. Routine verification shortens troubleshooting time and confirms changes actually took effect.

Port Security Configuration

```
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation shutdown
SW1(config-if)#switchport port-security mac-address 68b5.9965.1193
```

Why this is done / Benefit: Port security limits which and how many MAC addresses can use an access port. Restricting a port to one approved device helps prevent unauthorized endpoint swaps, unmanaged switches, and casual access. Shutdown mode places the interface into an error-disabled state when a violation occurs, making the event visible and contained.

Best-practice note: Hard-coded MAC addresses add administrative overhead. Sticky learning or 802.1X may be more scalable. Plan recovery and monitoring for error-disabled ports.

Port Security Verification

```
SW1#show mac-address-table
SW1#show port-security
SW1#show port-security interface fa0/5
```

Why this is done / Benefit: These commands show learned MAC addresses, global port-security state, violations, configured limits, and detailed status for a specific interface. They help determine whether the correct endpoint is connected and why a port may have shut down.

VLAN Setup

```
SW1(config)#vlan 10
SW1(config-vlan)#name SALES
SW1(config)#interface fastEthernet 0/5
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 10
```

Why this is done / Benefit: Creating and naming a VLAN provides logical segmentation independent of physical location. Assigning an access port to VLAN 10 places the connected endpoint in the SALES broadcast domain. Segmentation improves security, reduces broadcast scope, and makes policy, addressing, and troubleshooting easier to manage.

5. Trunking, VTP, and STP Optimization

Voice and Data VLANs

```
SW1(config)#interface fastEthernet 0/5
SW1(config-if)#switchport access vlan 16
SW1(config-if)#switchport voice vlan 12
```

Why this is done / Benefit: A data VLAN carries workstation traffic while a separate voice VLAN carries IP-phone traffic on the same switch port. Separation allows distinct addressing, QoS, security policy, and troubleshooting for voice and data while minimizing cabling requirements.

Best-practice note: Ensure the phone, switch, DHCP options, QoS policy, and upstream trunks all support the voice VLAN design.

Trunk Configuration

```
SW1(config)#interface fastEthernet 0/1
SW1(config-if)#switchport mode trunk
SW1(config-if)#switchport trunk allowed vlan add 10
```

Why this is done / Benefit: A trunk carries traffic for multiple VLANs between switches, routers, firewalls, or virtualization hosts. Restricting the allowed VLAN list reduces unnecessary broadcast propagation and limits which VLANs can traverse the link, improving both security and operational clarity.

Best-practice note: Confirm the native VLAN and allowed VLAN list on both ends. “add” appends VLAN 10 to the existing list; omitting “add” can replace the list on some platforms.

Trunk Hardening

```
SW1(config-if)#shutdown
SW1(config-if)#nonegotiate
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan 222
```

Why this is done / Benefit: Unused or untrusted ports should not negotiate trunks. Shutting the port down, disabling DTP negotiation, forcing access mode, and placing it in an unused VLAN reduces the risk of VLAN-hopping attacks, accidental loops, and unauthorized network access.

Best-practice note: The source heading says trunk hardening, but this block is really unused/access-port hardening. Apply it to unused ports or ports that must never become trunks.

VTP Configuration

```
SW1(config)#vtp mode server
SW1(config)#vtp domain EXAMPLE
SW1(config)#vtp password cisco
SW1(config)#vtp pruning
SW1(config)#vtp version 2
```

Why this is done / Benefit: VTP can distribute VLAN database changes across switches in the same domain, reducing manual configuration. A domain and password limit participation, and pruning reduces unnecessary flooded traffic on trunks that do not need a VLAN.

Best-practice note: VTP can propagate destructive VLAN changes if misconfigured. Many organizations use VTP transparent/off mode or VTPv3 with strict controls. Never deploy a switch without checking its VTP domain, mode, and revision behavior.

STP Root Bridge and Modes

```
SW1(config)#spanning-tree vlan 1 root primary
SW1(config)#spanning-tree vlan 1 root secondary
SW1(config)#spanning-tree vlan 1 priority 8192
SW1(config)#spanning-tree mode rapid-pvst
```

Why this is done / Benefit: Spanning Tree prevents Layer 2 loops. Intentionally selecting primary and secondary root bridges keeps traffic paths predictable and avoids an unsuitable access switch becoming root. Rapid PVST+ converges faster than classic 802.1D STP after topology changes.

Best-practice note: A switch should not normally be configured as both primary and secondary for the same VLAN. Use primary on the preferred root and secondary on a different backup switch.

STP Features and EtherChannel

```
SW1(config-if)#spanning-tree portfast
SW1(config-if)#spanning-tree bpduguard enable
SW1(config-if)#spanning-tree vlan 1 cost 25
SW1(config-if)#channel-group 1 mode on
```

Why this is done / Benefit: PortFast allows trusted edge ports to begin forwarding quickly, improving endpoint startup. BPDU Guard shuts an edge port if it receives a BPDU, helping prevent accidental loops or rogue switches. STP cost influences the preferred Layer 2 path. EtherChannel combines links for more bandwidth and redundancy while STP treats the bundle as one logical link.

Best-practice note: Use PortFast and BPDU Guard only on genuine edge ports. “mode on” creates a static EtherChannel and must match on both ends; LACP active/passive is often safer because it negotiates compatibility.

STP and VTP Verification

```
SW1#show interfaces switchport
SW1#show interfaces trunk
SW1#show vlan brief
SW1#show vtp status
SW1#show vtp password
```

Why this is done / Benefit: These commands confirm access/trunk mode, native and allowed VLANs, VLAN membership, VTP domain/mode/version, and whether a VTP password is configured. Verification prevents mismatched trunks, missing VLANs, and unexpected VTP behavior.

Best-practice note: Some IOS versions intentionally hide or limit display of the actual VTP password.

6. STP and CDP Troubleshooting

CDP Configuration

```
SW1(config)#cdp run
SW1(config-if)#no cdp enable
```

Why this is done / Benefit: CDP globally discovers directly connected Cisco devices and reports useful identity, platform, port, IP, and capability information. Disabling CDP on an individual

interface reduces information exposure on untrusted or external-facing links while retaining it internally for troubleshooting.

Best-practice note: CDP is Cisco proprietary. LLDP can provide standards-based discovery. Disable discovery protocols on ISP, guest, or otherwise untrusted links unless specifically required.

CDP Verification

```
SW1#show cdp
SW1#show cdp interface
SW1#show cdp neighbors
SW1#show cdp neighbors detail
SW1#show cdp entry *
```

Why this is done / Benefit: These commands verify whether CDP is running, identify participating interfaces, list directly connected neighbors, and display detailed neighbor information such as management IP, software, platform, and local/remote ports. This is highly useful for tracing cabling, validating topology, and locating misconnected devices.

Production Deployment Checklist

- Use unique strong secrets and AAA; do not deploy the example password “cisco.”
- Allow SSH only; remove Telnet and generate modern cryptographic keys.
- Use a dedicated management VLAN and restrict management access with ACLs.
- Document interfaces and verify running versus startup configuration after changes.
- Confirm VLAN, trunk, STP root, EtherChannel, routing, and authentication settings on both ends.
- Test changes during an approved maintenance window and maintain a rollback plan.
- Monitor logs, authentication failures, port-security violations, STP changes, and routing adjacencies.